TinyDistill: Knowledge Distillation into Compact Language Models

Mei Tanaka

Kestrel AI Lab

Workshop on Efficient Language Models (WELM), 2023

Abstract

TinyDistill studies Knowledge Distillation for Model Compression into a Small Language Model. A 70B teacher generates synthetic question-answer pairs and intermediate targets, while a 1.3B student learns both final answers and layer-wise representations. On Natural Questions, the student retains roughly 85% of teacher accuracy with 50x lower inference latency. The method is designed for Open-domain QA and focuses on preserving useful behavior under a strict parameter budget. It does not rely on external structured evidence; retrieval-augmented generation [1] is left as complementary future work.

Keywords

Knowledge Distillation, Model Compression, Natural Questions, Tiny models

1. Introduction

Large models can answer many questions but are expensive to serve. Model Compression seeks a smaller system that preserves task behavior, especially for Open-domain QA. TinyDistill uses Knowledge Distillation to transfer a teacher's answer distribution and intermediate signals into a Small Language Model. Retrieval-augmented generation [1] is a complementary way to add knowledge and is outside the present study.

2. Method

A 70B teacher produces synthetic question-answer pairs from Natural Questions prompts and supplies softened token distributions. The student is trained with a weighted combination of answer loss, distribution matching, and layer-wise representation alignment. A temperature schedule gives more weight to answer supervision late in training. To stabilize the Small Language Model, examples are filtered for duplicate answers and inconsistent spans. The resulting checkpoint is evaluated at several sequence lengths and batch sizes. We report Exact Match, parameter count, and latency on one accelerator, treating Model Compression as a practical deployment objective rather than only a score-matching problem.

3. Experiments

The 1.3B student reaches 56.1 EM on Natural Questions versus 66.0 for the teacher, retaining 85.0% of teacher accuracy. Median latency is 18 ms for the student and 910 ms for the teacher, a 50.6x reduction. Removing layer-wise supervision lowers EM by 2.7 points. The student remains weakest on rare entities and long answers.

System	Natural Questions EM	Latency	Parameters
70B teacher	66.0	910 ms	70B
TinyDistill student	56.1	18 ms	1.3B

4. Conclusion

TinyDistill shows that Knowledge Distillation can make Open-domain QA substantially cheaper. The Small Language Model retains most teacher accuracy while reducing latency by about fifty times. Retrieval is a promising next step for the knowledge gaps that distillation alone cannot close.

References

[1] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).